

Phosphorus

15 P Phosphorus is the most reactive member of group 15—white phosphorus ignites spontaneously in air. Phosphorus also plays a vital role in human biology. Like calcium, phosphorus strengthens bones and teeth. It is also present in deoxyribonucleic acid (DNA)—the chemical in our cells that carries our genetic information.



Apatite

This phosphorus-rich ore gets its name from the Greek word *apate*, which means “deceit,” because apatite often resembles crystals of other minerals, such as aquamarine.

Taste matters

A 12 oz (350 ml) can of cola contains about 0.002 oz (60 mg) of phosphorus, in the safe form of phosphoric acid. This ingredient is widely used in soft drinks to bring out a tangy, sharp taste. It also prevents the growth of bacteria in the drink.

Glass of cola

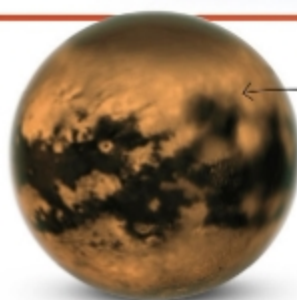


It's a trap!

All plants need nitrogen in large quantities to grow. The Venus flytrap grows in nitrogen-poor soil, so it supplements its nitrogen needs by feeding on insects and absorbing the element from them. When insects land on its pressure-sensitive leaves, they close to form a stomachlike space. This trap then fills with liquid chemicals that dissolve the prey, which is slowly absorbed through the surface of the leaves.

Venus flytrap

Flies supply 75 percent of the nitrogen that this plant needs to survive.



Titan's nitrogen-rich atmosphere gets its orange-brown color from hydrocarbon particles.

Nitrogen-rich atmosphere

Saturn's largest moon, Titan, is unusual because it's thought to be the only moon in the Solar System with a substantial atmosphere. Its atmosphere is about 95 percent nitrogen, and the rest is mostly methane. Most scientists think that nitrogen may have been delivered to Titan's atmosphere by ancient comets that crashed into it.

Nitrogen-based fuel is twice as powerful as gasoline.



Fueling the race

Nitromethane is an immensely powerful fuel made up of carbon, hydrogen, and nitrogen. This fuel generates the force that accelerates dragsters to speeds of more than 300 mph (480 kph).